

Superslip 6515AL & Superslip 6515AL-EZ

Introduction

Superslip 6516AL is a micronized wax based on high density polyethylene (HDPE), fatty acid amide wax and aluminum oxide. This unique composite powder utilizes nano aluminum oxide for its hardness to dramatically improve scratch performance. The HDPE and amide wax components provide superior slip and lubricity. Superslip 6515AL does not contain PTFE.

Superslip 6515AL can be difficult to disperse into water-based systems. Superslip 6515AL-EZ is the identical powder product but with a surface treatment to facilitate easy wetting and dispersion of the powder. Other than ease of dispersion, the performance benefits of Superslip 6515AL-EZ will be identical to Superslip 6515AL.

Superslip 6515AL can be compared to Polyfluo 150:

Property:	Superslip 6515AL	Polyfluo 150
Chemistry	HDPE/amide/alumina	HDPE/PTFE
Mean particle size (µm)	3.50-5.50	3.0 – 5.50
Maximum particle size (µm)	15.56	15.56
Melting point (°C)	139-145	113 – 116
Density at 25 °C (g/cc)	0.99	1.15

Objective

Compare the level of COF, abrasion, and scratch resistance between Superslip 6515AL and Polyfluo 150.

Test Methods

- ASTM D4060 Standard Test Method for Abrasion Resistance
- ASTM D3363 Standard Test Method for Film Hardness by Pencil Test (Scratch)
- ASTM D1894 Standard Test Method for Coefficient of Friction

Formulations and Study Methodology

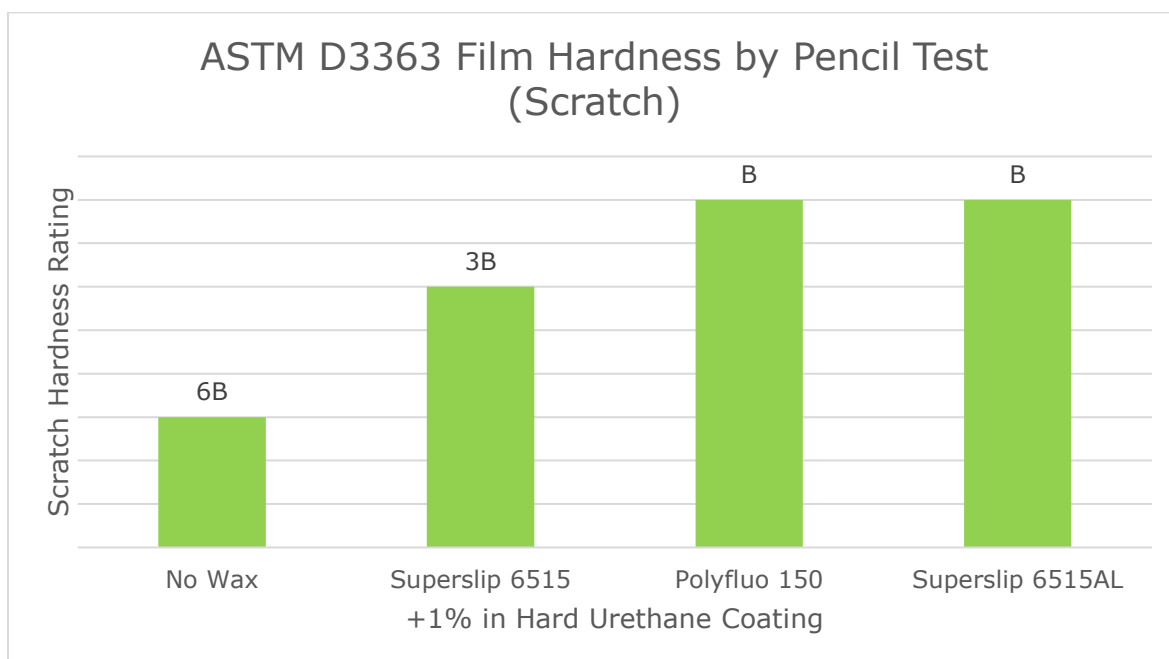
Hard Urethane Coating Formula:	Weight %:
Urotuf L63-MPW-38 (Reichhold)	97.5
Diethylene glycol monoethyl ether	2.0
BYK-345	0.2

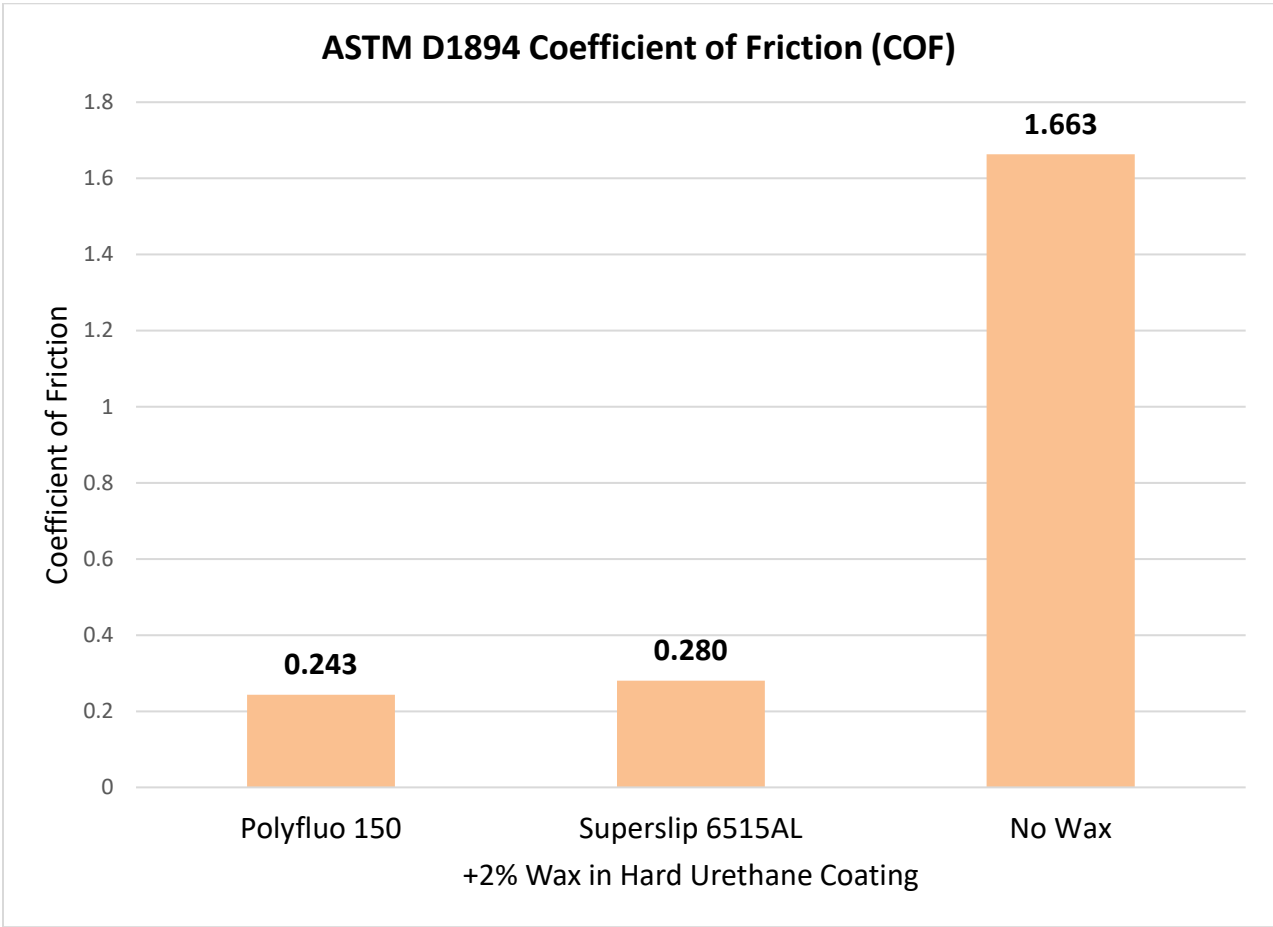
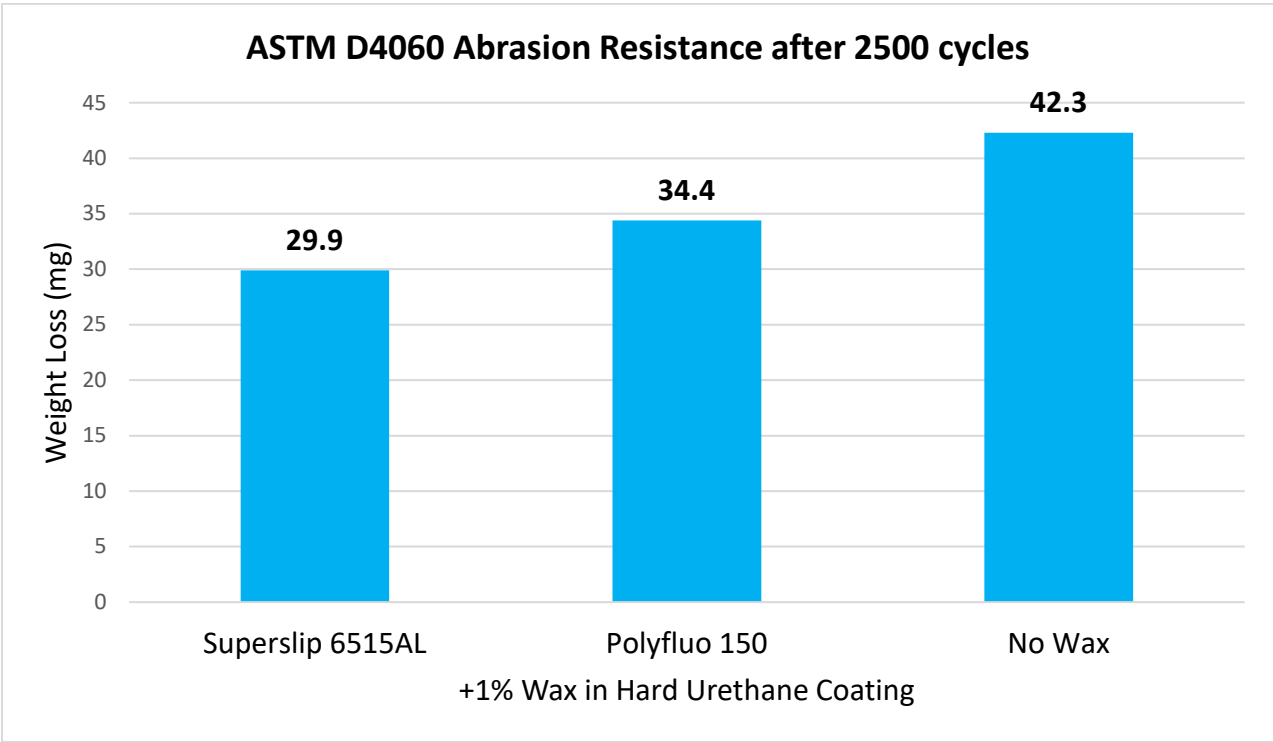
TEGO Foamex 805 N	0.3
Total:	100.0
Nonvolatiles (Solids):	38.4

Sample Preparation

- For the COF test, coatings were drawn down with a 3 mil (75 µm) Bird Film Applicator to produce approximately 1 mil (25 µm) DFT coatings.
- The coatings for the Taber Abrasion test were drawn down with a #60 wire wound rod to produce approximately 2 mil DFT coatings.
- The coatings for the Pencil Scratch test were drawn down with a #28 wire wound rod to produce approximately 1 mil DFT coatings.
- All coatings were allowed to cure for 7 days at room temperature at ~50% humidity.

Results





Conclusions

Superslip 6515AL compares favorably to several other products:

- **Comparable scratch resistance** vs. Polyfluo 150 **without the use of PTFE**
- **Improved scratch resistance** vs. Superslip 6515
- **10% improved abrasion resistance** vs. Polyfluo 150
- Comparable coefficient of friction to Polyfluo 150

Superslip 6515AL is an ultrafine PTFE-free composite wax additive recommended as a replacement for PTFE based waxes in can, coil and other thin film coating applications. If ease of dispersion into water-based systems is critical, Superslip 6515AL-EZ is recommended.